

Yiyan Liao

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EDUCATION

School of Life Sciences, Peking University

Sep. 2023 – Jun. 2027 (Expected)

Bachelor of Engineering in Bioinformatics

- Comprehensive Assessment Ranking: 7/134; GPA: 3.75/4.00
- Selected Coursework: Linux-based Essential Bioinformatics (A); Innovative Experiments in Bioinformatics (A); Bioinformatics (A); Artificial Intelligence for Protein Design (94/100); Computational Systems Biology (93/100)

SELECTED PUBLICATIONS

*Equal contribution. †Corresponding author.

[1] **Guiding Denoising Diffusion Models for Structure-based Drug Design via Machine-learned Interatomic Potentials**

Yiyan Liao*, Amin Tavakoli, Yu Zhang, Anima Anandkumar†

Submitted to the 2nd SIMBIOCHEM Workshop at NeurIPS 2026.

[2] **Molexar: A Unified Multimodal Molecular Foundation Model for Drug Design**

Haoyu Lin*, Yiyan Liao*, Jinmei Pan, Xinliao Ling, Luhua Lai†, Jianfeng Pei†

arXiv preprint, 2026. <https://doi.org/10.48550/arXiv.2606.25865>

[3] **MGTbind: a comprehensive database of molecular glue ternary interactome**

Jintao Zhu*, Yiyan Liao*, Haoyu Lin*, ..., Luhua Lai†, Jianfeng Pei†

Nucleic Acids Res., 54(D1), D1500–D1509, 2026. <https://doi.org/10.1093/nar/gkaf1075>

[4] **Benchmarking Cofolding Methods for Molecular Glue Ternary Structure Prediction**

Yiyan Liao*, Jintao Zhu*,†, Juan Xie, Luhua Lai†, Jianfeng Pei†

J. Chem. Inf. Model., 65(20), 11136–11148, 2025. <https://doi.org/10.1021/acs.jcim.5c01860>

RESEARCH EXPERIENCES

Anima AI + Science Lab, California Institute of Technology

Pasadena, CA

Summer Undergraduate Research Fellow (SURF) — Advisor: Prof. Anima Anandkumar

Mar. 2026 – Present

Guiding Denoising Diffusion Models for Structure-Based Drug Design via Machine-Learned Interatomic Potentials

- Guided structure-based drug design diffusion sampling with machine-learned interatomic potential forces to generate physically valid small-molecule drugs.
- Reduced protein-ligand collisions to zero on CrossDocked2020 and attained the best binding affinity scores.
- Ongoing: extending the guidance framework to autoregressive and flow-matching generative models.

Molecular Design Laboratory, Peking University

Beijing, China

Undergraduate Research Assistant — Advisors: Dr. Jianfeng Pei and Prof. Luhua Lai

Sep. 2024 – Present

MGTbind: a Comprehensive Database of Molecular Glue Ternary Interactome (Co-first author, Nucleic Acids Res.)

- Curated 3,093 molecular glue ternary complex entries from 227 publications, covering structures and activities.
- Modeled 3,607 complexes with AlphaFold 3 and annotated physicochemical descriptors and interactions.

Benchmarking Cofolding Methods for Molecular Glue Ternary Structure Prediction (First author, J. Chem. Inf. Model.)

- Built MGBench, a molecular glue ternary complex benchmark, from 221 curated structures with time-based splits isolating 88 unseen targets.
- Benchmarked 5 SOTA cofolding models including AlphaFold 3, showing their successes stem from memorization rather than atomic-level rules, with the largest gaps on novel E3 ligase systems.

DiffTernary: a Diffusion Model for Molecular Glue Ternary Complex Docking (Ongoing)

- Developed an SE(3)-equivariant diffusion model that predicts ternary docking poses by refining rigid-body placement and ligand torsions.
- Achieved a 10% higher acceptable rate of DockQ compared to the current SOTA method DeepTernary.

Molexar: a Unified Multimodal Molecular Foundation Model for Drug Design (Co-first author, preprint)

- Co-developed the Fragment-SELFIES molecular language and a Gemma2-based autoregressive decoder

conditioned on properties, pharmacophores, sequences, and binding pockets.

- Led evaluation against 20+ baselines, attaining 100% validity and SOTA results on the CrossDocked2020 test set and MolGenBench.
- Generated orthosteric hits for DPP4 and ACOD1 that outscore known inhibitors in Glide and MM-GBSA. Ongoing: wet-lab activity assays.

Peking University International Genetically Engineered Machine (iGEM) Team

Beijing, China

Wet Lab Leader — Advisor: Dr. Long Qian

Jan. 2024 – Oct. 2024

General Disease Detection with Protease and Aptamer (2024 iGEM Gold Medal)

- Measured binding affinities between aptamers and thrombin using Surface Plasmon Resonance (SPR) assays.
- Directed experimental strategy for a 15-member team and presented the final project defense at the Grand Jamboree in Paris.

PROFESSIONAL EXPERIENCES

Tencent

Beijing, China

Biological Large Language Model Intern

Jun. 2025 – Dec. 2025

- Developed a six-stage multi-agent system with retrieval-augmented generation and NCBI tools for automated *de novo* plasmid design, raising benchmark accuracy from 10% to 51%.

Moonshot AI

Beijing, China

Large Language Models (LLMs) Reinforcement Learning Intern

Jan. 2025 – Jun. 2025

- Built biology-domain reinforcement-learning data and ran pilot experiments for Kimi K2 on the Graduate-Level Google-Proof Q&A (GPQA) benchmark, informing the full-scale training runs.

SELECTED HONORS & AWARDS

Caltech Summer Undergraduate Research Fellowship (\$8,110 Grant)	2026
School of Life Sciences 87 Excellent Research Scholarship, First Prize (1 out of 10 finalists)	2026
Peking University Qin Wanshun Jin Yunhui Scholarship (top 10% of the cohort)	2025
Beijing Natural Science Foundation Undergraduate Research Program (\$7,000 Grant)	2025
International Genetically Engineered Machine (iGEM) 2024 Gold Medal (Wet Lab Leader)	2024
34th International Biology Olympiad (IBO) Gold Medal (3rd out of 293)	2023

ACADEMIC ACTIVITIES & SERVICE

Teaching Assistant of Biochemistry Laboratory, Peking University

Sep. 2025 – Present

- Instructed 25 students per semester in weekly biochemistry laboratory sessions.
- Established an *in silico de novo* design workflow for the GST enzyme using the protein language model ESM3.
- Awarded the Outstanding Teaching Assistant Award, School of Life Sciences, Peking University (Jun. 2026).

Student Advisor of Peking University iGEM Team 2025

Jan. 2025 – Oct. 2025

- Selected members and guided research for Team Peking 2025, which won a Gold Medal, the Best Model Award, and a Global Top 10 ranking.

Member of Undergraduate Honors Program in Biology (UHPB), Peking University

Sep. 2024 – Present

- Gave 4 journal club talks (1 hour each) and attended 16 more, once presenting to the senior author of the paper.
- Delivered an English talk on the molecular glue computational landscape to 100+ peers and 10 professors, including international faculty, at the 2025 UHPB Annual Symposium.

SKILLS & OTHER INFO

- **Programming Languages and Software:** Python, PyTorch, PyTorch Geometric, R, Bash, Git, Linux, SLURM, Schrödinger, GROMACS, Rosetta, AutoDock Vina, AlphaFold 3, ESM3, LLaMA Factory, PyMOL, ChimeraX
- **Laboratory Equipment and Skills:** Molecular Cloning, Site-Directed Mutagenesis, Protein Expression and Purification, ÄKTA HPLC, SPR, EMSA, SDS-PAGE, Western Blot, Co-IP, PCR/qPCR, Bacterial Culture
- **Languages:** TOEFL: 103/120